

Alina Guzu

Data Scientist

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Data Scientist with 3+ years of experience in medical imaging, machine learning, and AI applications for healthcare. Experienced in Python programming, image analysis, predictive modeling, and developing data workflows. Motivated problem-solver with strong communication skills and a research-oriented mindset. Dedicated to transforming complex data into actionable insights through deep learning, data structuring, and automation. **Based in San Diego, does not require visa sponsorship.**

Key Qualifications

- Strong analytical skills with proven experience interpreting complex medical and research data
- Experienced in MRI/CT image segmentation, deep learning-based classification, and predictive modeling
- Skilled in AWS cloud solutions and SaaS-based software implementation for laboratory environments
- Strong cross-functional communication and project coordination across scientific, technical, and clinical teams
- Adaptable, fast learner, and committed to continuous improvement in data-driven environments

Education

2024 **Utrecht University, Master's in Biomedical Engineering, Utrecht, The Netherlands**

- Specialization in Medical Imaging
- 2022 - Completed Honours Programme focusing on interdisciplinary seminars and collaborative projects
 - Developed skills in communication, perspective-taking, and integration across disciplines
 - Produced deliverables such as presentations, essays, and podcasts

2018 **Maastricht University, Bachelors of Science in Mathematics, Economics, Data Science,, Maastricht, The Netherlands**

- Personalized curriculum as part of the Bachelor's Honours Programme in Liberal Arts and Sciences
- Problem-Based Learning

Employment History

June 2025 – **Lab Specialist Project Manager, AgileBio, San Diego**

- Present
 - Lead implementation projects of Laboratory Information Management Systems
 - Coordinate between software developers, laboratory staff, and stakeholders
 - Translate scientific problems into practical solutions
 - Manage project timelines and resources while providing end-user training and support to ensure successful LIMS adoption and usability

May 2025 – **Volunteer, Red Cross, San Diego**

- Present
 - Outreach volunteer for the WIC (Women, Infants and Children) program

May 2022 – **Research Assistant and Intern Diffusion MRI, University Medical Centre Utrecht, The Netherlands**

Jan 2025 ○ Jun 2023 – Jan 2025 **Research Assistant**

- Contributed to a high-impact project predicting the risk of intracranial aneurysm development
- Conducted detailed annotation and analysis of MRA images utilizing MeVisLab

○ May 2022 – Mar 2023 **Intern Diffusion MRI**

- Evaluated the potential application of Intra-Voxel Incoherent Motion (IVIM) to Spherical Tensor Encoding (STE) data, comparing it to conventional diffusion MRI for quantifying blood and water diffusion in tissues
- Conducted detailed analysis using MATLAB

Sept 2023 – **Intern Biomedical Engineering & Physics, University Medical Centre Amsterdam**

- Feb 2024
 - Developed an AI-based tool to enhance body composition analysis in abdominal CT scans
 - Collaborated with a radiologist to obtain large datasets
 - Achieved high accuracy in body component segmentation with dice scores above 0.9
 - The tool is currently used in Amsterdam for cancer patient care

Mar 2021 – **Microbiologist, Balans (Specialized in laboratory, process engineering, and executive recruitment), Rotterdam, The Netherlands**

- Jun 2021
 - Conducted COVID-19 testing using PCR techniques, including primer and PCR Master Mix preparation, machine operation, and result determination
 - Performed routine lab tasks, ensuring accurate sample processing, timely testing, and adherence to safety protocols during the pandemic

- Dec 2020 – **Volunteer**, *Ikazia Hospital*, Rotterdam, The Netherlands
- Mar 2021 – Assisted in patient transportation, including guiding patients to their appointments and surgeries
- Jun 2019 – **Agile Test Software Engineer**, *Vodafone*, Maastricht, The Netherlands
- Aug 2020
 - Project Vodafone Heerlen: Network Tester
 - Tested all forms of calling and conducted extensive testing on 5G, 4G, and emergency call systems
 - Vodafone Ziggo Maastricht: Software Tester
 - Assisted in manual and automated testing of Vodafone's new website, enhancing user experience and functionality
 - Various Diplomas and courses obtained:
 - Business summer course, **Ohio University, USA** (July 2019 - August 2019)
 - Test Automation Tosca Certificate, TMap Next Diploma, SCRUM Master diploma
- Sep 2018 – **IT Intern**, *Medtronic*, Maastricht
- Jan 2019
 - Contributed to the development plan for a global mobile app to access employee contact information securely
 - Documented privacy requirements and collaborated with managers and employees across various teams to gather feedback and align on app functionalities
 - Conducted interviews with staff to understand the app's needs and ensure it met organizational standards

Technical Skills

- **Programming Languages:** Python, MATLAB, C++, Java, SQL, Bash
- **Technologies & Frameworks:** TensorFlow, PyTorch, OpenCV, scikit-learn
- **Cloud & SaaS:** AWS, SaaS deployment, Data migration
- **Tools/Software:** MeVisLab, Tableau, PowerBI, SPSS, Weka, Git/GitHub, Jupyter Notebooks, Microsoft Excel, Overleaf (LaTeX), Mendeley, BLAST
- **Medical Physics:** MRI and CT image analysis, Ultrasound physics, Radiotherapy dose optimization, DICOM, Segmentation Tools (MeVisLab, ITK-SNAP)
- **Data Science:** Data Analysis, Deep Learning, Predictive Modeling, Image Processing

Languages

Romanian: Native **English:** Fluent **Dutch:** Fluent **Spanish:** Intermediate **French:** Intermediate